

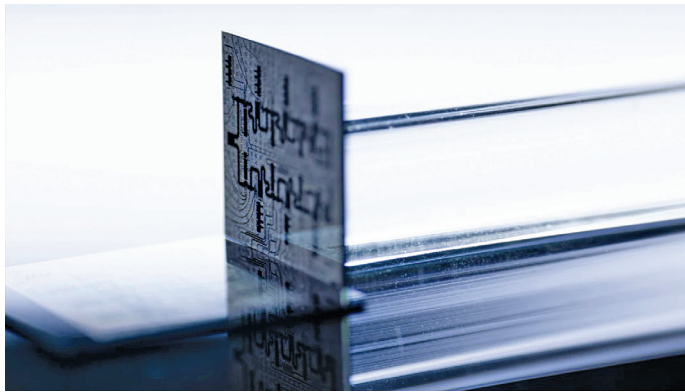
200 seconds for a task that would have taken a state-of-the-art supercomputer 10,000 years to complete. This clearly established the unique capabilities of quantum vis-a-vis classical computing.

In December 2024, Google showcased Willow, a 105-qubit quantum chip that excels in performance and error-correction, the second milestone on their roadmap.

Performance. Willow performed a random-circuit sampling (RCS) benchmark computation in under five minutes, which would have taken today's best supercomputers around 10 septillion (10^{25}) years to solve. Quantum benchmark tests help to determine whether a system qualifies as a quantum computer or whether it is slipping into the classical realm and how well it performs compared to the theoretically expected values in a noise-free quantum computer. RCS is one such quantum benchmarking system developed by Google, which involves executing a random quantum circuit (layers of gates) to generate output bitstrings, forming a high-dimensional probability distribution beyond the capabilities of classical simulation.

Error correction and other achievements. Error correction is one of the most important challenges in quantum computing. Usually, as the number of qubits increases, more errors occur, and the system slips to classical levels. However, as the number of qubits used in Willow increases, the errors reduce, and the system becomes more quantum.

In surface code quantum computing, qubits are grouped into square lattices called surface codes,



The pair of silicon microchips that compose the Ocelot logical-qubit memory chip (Courtesy: AWS)

each of which represents a single encoded or logical qubit. This is one of the popular techniques of error correction in quantum computing—basically, the data is protected by distributing it across multiple physical qubits. As the lattice gets bigger, the logical qubit is more protected, and it is expected to perform better. But a bigger lattice also means more opportunities for error. If the error rate of the physical qubits is too high, error correction will not work as expected, and the extra errors will make the processor's performance worse.

However, if the error rate of the physical qubits is below a critical threshold, quantum error correction becomes extremely effective, and even small improvements in

the device are amplified exponentially. That explains why the operation fidelities in Willow are only around twice as good as Sycamore, but the encoded error rates are about twenty times better.

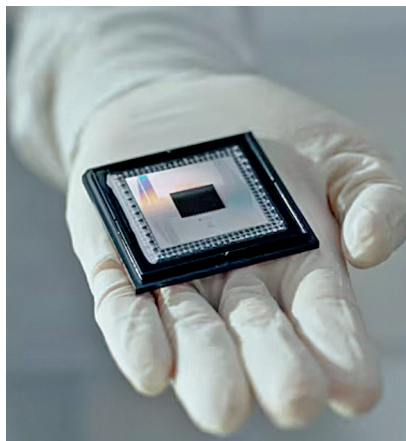
Technologists have been working towards going 'below the threshold' for nearly three decades—and Willow is reportedly the first ever

prototype to achieve this goal. Apart from qualifying as 'below threshold'—being able to drive errors down while scaling up the number of qubits—the demonstration also qualifies as 'beyond breakeven'—where arrays of qubits have longer lifetimes than individual physical qubits.

Willow maintains the tunability of the Sycamore architecture while improving the average qubit lifetimes (T_1) from about $20\mu\text{s}$ to $68\mu\text{s} \pm 13\mu\text{s}$. The qubits and operations in Willow are optimised for quantum error correction and run alongside their error correction software, which incorporates cutting-edge machine learning, reinforcement learning, and graph-based algorithms to identify and correct errors accurately.

"Each time we increase our lattice in size from 3x3 to 5x5 to 7x7, the encoded error rate decreases by a factor of 2.14. This culminates in a logical qubit whose lifetime is more than twice that of its best constituent physical qubit, demonstrating the capacity of an error-corrected qubit to go beyond its physical components," says a report written by Google research scientists Michael Newman and Kevin Satzinger.

Way ahead. So, can this error correction scale to the level of error rates required in a useful large



Google Quantum AI's quantum chip, Willow (Courtesy: Reuters)